JanGyan — Learn Without Limits

Category 6: Environment & Civic Responsibility

Module 6 – Climate Change & Disaster Awareness

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Climate Change & Disaster Awareness** — what climate change is, how it is already reshaping India's weather patterns, and the practical steps every citizen can take to build resilience and stay safe. Whether you are a student, a parent, a farmer, or a civic volunteer, this handbook is written so you can read it once and walk away confident and prepared.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of climate change and disaster risk reduction in India.

Learning Objectives

After completing this module, you will be able to:

- Explain climate change, global warming, and the greenhouse effect in simple terms
- Understand your own carbon footprint and how it is calculated
- Identify the major causes and effects of climate change in India
- Distinguish between climate mitigation and climate adaptation
- Apply basic disaster risk reduction and preparedness practices
- Understand what climate resilience means for households and communities
- Recognise India's key climate initiatives and the government bodies driving them
- Know exactly how and where to access disaster alerts and emergency support

Why This Topic Matters

Climate change is no longer a distant, future concern for India — it is reshaping weather patterns happening right now, from unseasonal heatwaves to erratic monsoons and more frequent flooding. What were once rare, once-in-a-century events are increasingly becoming part of ordinary annual patterns.

> ***Reality Check:*** *India recorded extreme weather events on 331 of 334 days between January and November 2025 — up from 295 days in 2024 and 292 days in 2022 — with at least 4,419 deaths reported in 2025 alone (Centre for Science and Environment, 'Climate India 2025'). Separately, the Union Home Ministry reported that over 2,900 people lost their lives to hydro-meteorological disasters in 2024–25.*

These are not distant statistics — they represent floods, heatwaves, cyclones, and landslides affecting real communities across nearly every state. Understanding both climate change and disaster preparedness has become essential knowledge for every Indian citizen, not just those working in environmental science or emergency response.

This is why JanGyan continues its Environment & Civic Responsibility category with this essential, practical module.

Introduction

Climate change refers to long-term shifts in temperature, precipitation, and weather patterns, primarily driven since the industrial era by human activities that increase greenhouse gas concentrations in the atmosphere. **Disaster awareness** is the knowledge and preparedness needed to reduce harm from the natural hazards that climate change is intensifying — floods, heatwaves, cyclones, and droughts.

In simple words: climate change is the slow shift in our planet's long-term weather patterns, while disaster awareness is what protects us in the shorter, sharper moments when those patterns produce extreme, dangerous events.

The two are deeply connected — as the climate changes, the frequency and intensity of extreme weather events tends to increase, which is exactly what India's own disaster data has shown over the past several years.

Fundamental Concepts

Before going deeper, let us understand a few foundational ideas that the entire subject of climate change is built upon.

The Greenhouse Effect

The greenhouse effect is a natural process where certain gases trap heat in the atmosphere, keeping Earth warm enough to sustain life. Human activity has intensified this effect far beyond its natural balance.

Greenhouse Gas	Major Human Source
Carbon Dioxide (CO2)	Burning fossil fuels, deforestation
Methane (CH4)	Agriculture, livestock, landfills
Nitrous Oxide (N2O)	Fertiliser use, industrial processes
Fluorinated Gases	Industrial refrigerants and manufacturing

Key Definitions

- **Global Warming** — The long-term rise in Earth's average surface temperature, primarily due to increased greenhouse gas emissions.
- **Carbon Footprint** — The total greenhouse gases generated by a person, product, or activity, usually measured in CO2 equivalent.
- **Mitigation** — Action taken to reduce or prevent the emission of greenhouse gases.
- **Adaptation** — Adjusting systems, infrastructure, and practices to cope with the impacts of climate change that are already unavoidable.
- **Disaster Risk Reduction (DRR)** — Systematic efforts to reduce vulnerability to hazards through preparedness, early warning, and resilient infrastructure.

Complete Detailed Explanation

Part A — India's Changing Weather Patterns, In Depth

According to the Centre for Science and Environment's annual assessments, extreme weather events in India have escalated sharply in recent years — occurring on 255 of 274 days in the first nine months of 2024, and on a striking 331 of 334 days between January and November 2025. The report also notes a collapse of clear seasonal boundaries, with coldwaves arriving earlier and warm nights becoming more frequent and severe.

These events are not evenly distributed — states like Himachal Pradesh, Kerala, Assam, and Andhra Pradesh have recorded particularly high numbers of disaster-affected days, while flood-prone, riverine states in the east and northeast face disproportionate impact due to their geography and population density.

Part B — The Ten Building Blocks of Module 6

This is the heart of the module. Each topic below is explained with its core meaning and its practical relevance for Indian citizens.

1. What is Climate Change?

Climate change refers to long-term shifts in temperature, precipitation, and weather patterns, largely driven since the industrial era by human activities that increase greenhouse gas concentrations in the atmosphere.

2. Global Warming

Global warming is the long-term rise in Earth's average surface temperature, primarily due to increased greenhouse gas emissions from burning fossil fuels, deforestation, and industrial activity. It is the central driver behind broader climate change.

3. Greenhouse Effect

The greenhouse effect is a natural process where gases like carbon dioxide and methane trap heat in the atmosphere, keeping Earth warm enough to sustain life. Human activity has intensified this effect, causing excess warming.

4. Carbon Footprint

A carbon footprint is the total amount of greenhouse gases generated by an individual, organisation, or activity, through actions like travel, energy use, and consumption habits — understanding it is the first step toward reducing it.

5. Causes & Effects of Climate Change

Causes include burning fossil fuels, deforestation, industrial emissions, and intensive agriculture; effects include rising temperatures, melting glaciers, more frequent extreme weather events, and disrupted ecosystems and biodiversity.

6. Climate Action

Climate action involves mitigation (reducing emissions through clean energy, afforestation, and efficiency) and adaptation (adjusting infrastructure and practices to cope with unavoidable climate impacts), at both policy and individual levels.

7. Disaster Risk Reduction

Disaster risk reduction (DRR) aims to minimise vulnerability to hazards like floods, cyclones, and heatwaves through early warning systems, resilient infrastructure, community preparedness, and sound land-use planning.

8. Climate Resilience

Climate resilience is the capacity of communities, economies, and ecosystems to anticipate, absorb, and recover from climate-related shocks and stresses, while adapting to long-term changes.

9. India's Climate Initiatives

National frameworks and commitments guide climate mitigation and adaptation efforts, including the National Action Plan on Climate Change, India's net-zero commitment for 2070, and the LiFE (Lifestyle for Environment) movement.

10. Sustainable Future

Building a sustainable, climate-resilient future requires integrating clean energy, disaster preparedness, resilient infrastructure, and inclusive policy-making — supported by informed and engaged citizens at every level of society.

Important Terminologies

Term	Simple Meaning
Adaptation	Adjusting systems and practices to cope with unavoidable climate impacts
Carbon Footprint	Total greenhouse gases generated by a person, product, or activity
Climate Resilience	The capacity to anticipate, absorb, and recover from climate shocks
Disaster Risk Reduction (DRR)	Systematic efforts to reduce vulnerability to hazards
Early Warning System	A system that detects and communicates hazard risk before it strikes
Extreme Weather Event	A weather occurrence significantly outside the normal historical range
Greenhouse Effect	The natural warming process caused by gases trapping heat in the atmosphere
Mitigation	Action taken to reduce or prevent greenhouse gas emissions
Net Zero	A state where greenhouse gas emissions are balanced by removals from the atmosphere
NDC (Nationally Determined Contribution)	A country's climate action targets submitted under the Paris Agreement

Step-by-Step Understanding

How Climate Change Intensifies Extreme Weather

1. Emissions — Human activity releases greenhouse gases into the atmosphere

2. Warming — These gases trap additional heat, raising global average temperatures

3. Disrupted Patterns — Warmer air and oceans alter rainfall, wind, and storm patterns

4. Intensified Events — Heatwaves, cyclones, and rainfall events become more frequent or severe

5. Compounding Impact — Communities face repeated, overlapping disasters with less time to recover between events

How to Prepare Your Household for a Climate-Related Disaster

1. Identify the specific hazards common to your region (floods, heatwaves, cyclones, landslides)

2. Keep an emergency kit with essentials: water, medicines, torch, important documents

3. Save official disaster alert numbers and apps on your phone

4. Know your nearest safe evacuation point or community shelter

5. Discuss and practise a basic family emergency plan

6. Stay informed through IMD and NDMA alerts during high-risk seasons

Real-Life Indian Examples

(Widely reported public events, described for educational awareness.)

- **2025's Record-Breaking Extremes:** The year saw multiple climate records broken — January was India's fifth driest since 1901, February was the warmest in 124 years, and September ranked as the seventh-highest mean temperature on record for that month (CSE, 'Climate India 2025').
- **Assam and Northeast Flooding:** Riverine states like Assam recorded some of the highest numbers of flood-related disaster days in 2025, displacing large numbers of people and damaging agricultural land.
- **Himachal Pradesh's Repeated Disasters:** The state recorded the highest number of extreme weather days nationally in late 2025, alongside the highest hydro-meteorological death toll (408 deaths in 2024–25) of any state, driven by cloudbursts, flash floods, and landslides.
- **Mission LiFE (Lifestyle for Environment):** A citizen-focused global initiative launched by India, encouraging individuals to adopt mindful, sustainable lifestyle choices as a collective climate action strategy.

Practical Scenarios

Scenario 1: Your region has received an official heatwave warning for the coming week.

Avoid outdoor activity during peak afternoon hours, stay hydrated, check on elderly neighbours and relatives, and know the early symptoms of heat exhaustion.

Scenario 2: Heavy, sustained rainfall is forecast for your area, and your home is in a low-lying zone.

Move important documents and valuables to higher ground in advance, keep an emergency kit ready, and follow local authority evacuation guidance if issued.

Scenario 3: Your community is planning new construction on what used to be a natural floodplain or wetland.

Raise concerns during public consultation — building on natural drainage areas increases flood risk for the surrounding community, not just the new development.

Government Rules / Guidelines

- **National Disaster Management Authority (NDMA):** The apex national body for disaster management in India, leading planning, mitigation, response, and recovery under the Disaster Management Act, 2005.
- **India Meteorological Department (IMD):** India's official weather forecasting agency, issuing warnings for cyclones, heatwaves, heavy rainfall, and other hazardous weather.
- **Ministry of Environment, Forest and Climate Change (MoEFCC):** Coordinates India's national climate policy, including the National Action Plan on Climate Change and India's NDC commitments.
- **State and District Disaster Management Authorities:** Implement disaster preparedness, response, and recovery at the state and local level, supported by NDMA guidelines.

Laws & Policies

Law / Policy	What It Covers
Disaster Management Act, 2005	Establishes the legal and institutional framework for disaster management in India, including the NDMA
National Action Plan on Climate Change (NAPCC)	Eight national missions covering solar energy, water, agriculture, habitat, and more
India's NDC under the Paris Agreement	Commits India to net-zero emissions by 2070 and specific interim 2030 targets
Mission LiFE (Lifestyle for Environment)	A global citizen movement led by India promoting mindful, sustainable individual lifestyle choices
State Disaster Management Plans	State-level frameworks aligning local disaster preparedness with the national policy
Environment (Protection) Act, 1986	The umbrella law empowering government action against activities that harm the environment

> *India's Nationally Determined Contribution (NDC) commits the country to achieving net-zero greenhouse gas emissions by 2070, alongside interim targets for renewable energy capacity and emissions intensity reduction by 2030.*

Important Statistics

Metric	Figure
Days with extreme weather events (Jan–Nov 2025)	331 of 334 days (99% of days)
Days with extreme weather events (2024, full year)	314 of 365 days
Deaths from extreme weather events in 2025	At least 4,419
Deaths from extreme weather events in 2024	Approximately 2,933–3,238 (source-dependent)
Deaths from hydro-meteorological disasters (2024–25, MHA)	Over 2,900
Highest state death toll (2024–25)	Himachal Pradesh — 408 deaths
People internally displaced by climate change in India (IDMC estimate)	About 14 million
India's net-zero emissions target year	2070

(Figures sourced from the Centre for Science and Environment (CSE) 'Climate India 2025' report, the Union Ministry of Home Affairs, and the Internal Displacement Monitoring Centre (IDMC); figures vary slightly by source and reporting period, so official portals should be checked for the latest updates.)

Best Practices

- Stay informed through official IMD and NDMA weather alerts, especially during high-risk seasons
- Keep a basic emergency kit ready at home (water, medicines, torch, documents)
- Reduce your personal carbon footprint through daily energy and consumption choices
- Support and participate in local disaster preparedness drills where available
- Discuss and practise a family emergency plan
- Advocate for climate-resilient infrastructure in your community

Common Mistakes

- Ignoring official heatwave, flood, or cyclone warnings
- Assuming climate change is a distant, future problem rather than a present reality
- Having no emergency plan or kit prepared before disaster season begins
- Building or supporting construction on natural floodplains and drainage areas
- Underestimating the cumulative impact of individual carbon footprints

Myths vs Facts

Myth	Fact
"Climate change is a slow process with no visible effects yet"	India recorded extreme weather on 99% of days between January and November 2025, showing measurable, present-day impact
"Only coastal areas need to worry about climate change"	Heatwaves, erratic rainfall, and flooding now affect nearly every region of India, including inland and hill states
"Individual actions don't matter for a global problem"	Collective individual and community action, alongside policy change, is part of both mitigation and building local resilience
"Disaster preparedness is only the government's job"	Household and community-level preparedness significantly reduces harm and complements official disaster response systems

Do's and Don'ts

Do	Don't
Follow official IMD and NDMA weather alerts	Don't ignore heatwave, flood, or cyclone warnings
Keep an emergency kit and family plan ready	Don't wait until a disaster strikes to prepare
Reduce your personal carbon footprint daily	Don't assume individual actions are insignificant
Support climate-resilient local infrastructure	Don't support construction on natural floodplains
Check on vulnerable neighbours during extremes	Don't overlook elderly or at-risk community members

Practical Tips

For Households

Keep an emergency kit with water, medicines, and a torch • Save NDMA and local disaster helpline numbers • Know your nearest safe shelter or evacuation point

For Communities

Participate in local disaster preparedness drills • Support climate-resilient drainage and green infrastructure • Set up neighbourhood check-in systems for vulnerable residents during extreme weather

For Farmers

Diversify crops to reduce climate-related risk • Use IMD agro-advisories for planting decisions • Invest in water-harvesting structures to buffer against erratic rainfall

Practical Checklist

■	I know the specific climate hazards common to my region
■	I have an emergency kit ready at home
■	I have saved official IMD/NDMA alert numbers or apps
■	I know my nearest safe evacuation point or shelter
■	I have discussed a family emergency plan with my household
■	I take steps to reduce my personal carbon footprint
■	I have discussed climate awareness and preparedness with family or peers

Quick Revision Notes

- Climate change = long-term shifts in temperature and weather patterns driven by greenhouse gas emissions
- Mitigation reduces emissions; adaptation adjusts to unavoidable climate impacts
- India recorded extreme weather on 99% of days between January and November 2025
- NDMA, IMD, and MoEFCC are India's key climate and disaster management bodies
- India's NDC targets net-zero emissions by 2070
- Disaster Management Act, 2005 establishes India's institutional disaster response framework

Key Takeaways

1. Climate change is not a distant future risk — India's own data shows measurable, escalating impact in the present day

2. Disaster preparedness at the household level significantly reduces harm and complements official emergency response

3. Mitigation and adaptation are both necessary — reducing emissions while also preparing for unavoidable impacts

4. Certain states and communities face disproportionate climate risk, requiring targeted resilience investment

5. Personal and collective action, from carbon footprint reduction to community preparedness, genuinely shapes outcomes

Glossary

- **Adaptation** — Adjusting systems and practices to cope with unavoidable climate impacts
- **Carbon Footprint** — Total greenhouse gases generated by a person, product, or activity
- **Climate Resilience** — The capacity to anticipate, absorb, and recover from climate shocks
- **Disaster Risk Reduction (DRR)** — Systematic efforts to reduce vulnerability to hazards
- **Extreme Weather Event** — A weather occurrence significantly outside the normal historical range
- **Global Warming** — The long-term rise in Earth's average surface temperature
- **Greenhouse Effect** — The natural warming process caused by gases trapping heat in the atmosphere
- **Mitigation** — Action taken to reduce or prevent greenhouse gas emissions
- **NDC** — Nationally Determined Contribution, a country's climate targets under the Paris Agreement
- **Net Zero** — A state where greenhouse gas emissions are balanced by removals from the atmosphere

Frequently Asked Questions

1. What is the difference between climate change and global warming?
Global warming refers specifically to the long-term rise in Earth's average temperature; climate change is the broader set of shifts in temperature, rainfall, and weather patterns that result from it.

2. What is the difference between mitigation and adaptation?
Mitigation means reducing or preventing greenhouse gas emissions at the source; adaptation means adjusting infrastructure and practices to cope with climate impacts that are already unavoidable.

3. How much has extreme weather increased in India recently?
India recorded extreme weather events on 331 of 334 days between January and November 2025 — up from 295 days in 2024 and 292 days in 2022, according to CSE's Climate India assessments.

4. What is a carbon footprint?
The total amount of greenhouse gases generated by a person, product, or activity, usually measured in units of CO2 equivalent.

5. What is India's net-zero target?
India has committed to achieving net-zero greenhouse gas emissions by 2070 under its Nationally Determined Contribution (NDC) to the Paris Agreement.

6. What is the National Disaster Management Authority (NDMA)?
India's apex body for disaster management, responsible for planning, mitigation, response, and recovery under the Disaster Management Act, 2005.

7. What is Mission LiFE?
A global citizen movement led by India, encouraging individuals to adopt mindful, sustainable lifestyle choices as a form of collective climate action.

8. Which Indian states face the highest climate disaster risk?
States like Himachal Pradesh, Kerala, Assam, and Andhra Pradesh have recorded particularly high numbers of disaster-affected days in recent assessments, though risk profiles vary by hazard type.

9. How can I prepare my household for a climate-related disaster?
Keep an emergency kit ready, save official alert numbers, know your nearest safe shelter, and discuss a basic family emergency plan in advance.

10. Where can I check official weather and disaster alerts?
The India Meteorological Department (IMD) issues official weather warnings, while the NDMA and state disaster management authorities provide broader disaster preparedness guidance.

Official References

- Centre for Science and Environment (CSE) / Down To Earth — ‘Climate India 2025: An Assessment of Extreme Weather Events’
- National Disaster Management Authority (NDMA) — Annual Report 2024–25
- Union Ministry of Home Affairs — hydro-meteorological disaster data
- Ministry of Environment, Forest and Climate Change (MoEFCC) — National Action Plan on Climate Change and India's NDC
- Internal Displacement Monitoring Centre (IDMC) — climate-related displacement estimates for India
- Disaster Management Act, 2005 — India Code portal

Official Government Website Links

- National Disaster Management Authority: **https://ndma.gov.in**
- India Meteorological Department: **https://mausam.imd.gov.in**
- Ministry of Environment, Forest and Climate Change: **https://moefcc.gov.in**
- Mission LiFE: **https://missionlife-moefcc.nic.in**
- Press Information Bureau: **https://pib.gov.in**

Conclusion

Climate change is no longer a topic for the future — it is visible in India's weather every single week. But understanding it, preparing for it, and reducing our own contribution to it are all within reach for every citizen. Resilience is not built in the moment of a disaster; it is built in the ordinary days before it, through preparation, awareness, and daily habits.

Continue your Environment & Civic Responsibility journey with JanGyan. Stay alert, stay prepared, and remember: the scale of the challenge asks for the scale of our response.

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